14.02 - Principles of Macroeconomics
Problem Set 4

Spring 2023

Question 1 - The WS-PS Model [20 Points]

Suppose the wage-setting equation is given by:

$$W = P\,(1-u)\,z$$

where z represents unemployment insurance, u represents the unemployment rate and W represents the nominal wage . Suppose the production function is given by $Y = N$, where N is employment, and the price-setting equation is given by:

$$P = W\,(1+m)$$

where m is the markup.

1. [6 Points] Explain how the wage-setting equation depends on the price level, the unemployment rate, and the unemployment insurance. Provide the intuition.

   1. price level: the reason it depends on the price level is because the wage is determined by the mark-up so the higher the price the higher the wage, moreover higher price gives more room for workers to demand higher wages 2. unemployment: they have a negative relationship because higher unemployment gives lower bargaining power to the workers. 3. the unemployment insurance: because it raises their opportunity cost it means they demand higher wages to stay in their position.
2. [6 Points] Explain how the price-setting equation depends on the wage and the markup m. Provide the intuition.

   firstly the cost of an extra unit (the MC) is the wage here so that is the smallest P they can charge, moreover the markup is their profit on each unit, and is possitively correlated with the P, and is dependent on the market environment.

3. [8 Points] Express both the wage-setting equation and the price-setting equation in terms of the real wage W/P. Find algebraically the natural rate of unemployment, u_n, by equating the real wage implied by the wage-setting equation with the one implied by the price-setting equation. Does the natural rate of unemployment increase or decrease in the mark-up? Provide intution.

   $\frac{W}{P} = (1-u)z$ $\quad \frac{W}{P} = \frac{1}{1+m}$ $\quad (1-u^n)\,z = \frac{1}{1+m}$ $\quad -u^n = \frac{1}{(1+m)z} - 1$ $\quad u^n = 1 - \frac{1}{z(1+m)}$

   u^n increases in the markup. because with $\bar{P}$ the higher the m then w and thus u needs to adjust.

Question 2 - The Phillips Curve (30 points)

Consider the WS and PS relations in a dynamic setting where the subscript t represents the year:

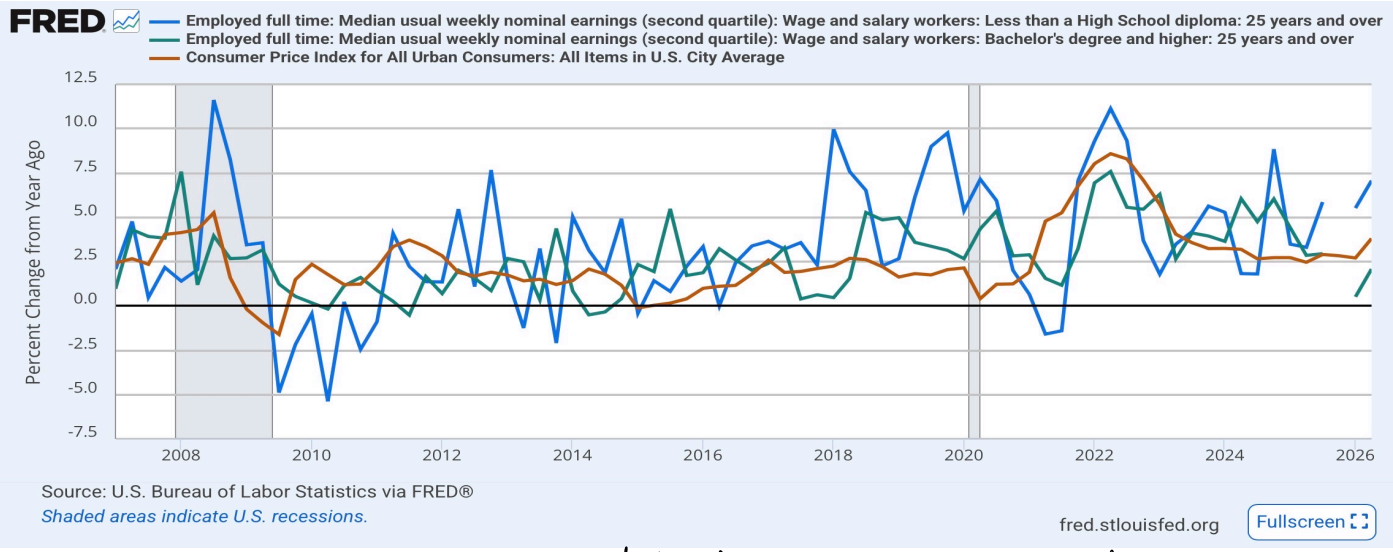
$$W_t = P_t^e F(u_t, z)$$
$$P_t = (1+m) W_t$$

We shall assume for this question that:

$$F(u_t, z) = 1 - \alpha u_t + z$$

where α captures the strength of the effect of the unemployment rate on the equilibrium wage, and z represents the catch-all variable that captures the effect on wages of the policy environment (e.g., minimum wage, employment protection legislation, etc).

1. [8 points] Go to the FRED website (link here). Download the data series for median usual weekly earnings for those with less than a high school degree and those with a bachelor's degree and higher (series LEU0252916700Q and series LEU0252918500Q). Also download the data series for the Consumer Price Index for All Urban Consumers: All Items in US City Average (series CPIAUCSL) since 2007. Calculate the percent change from a year ago at a quarterly frequency for each series and plot them together in one chart. Calculate the average for each of these three series from 2007 to 2021. Using these averages, what can you say about wage inflation and price inflation in the last 15 years? Is there a difference across educational groups? What has happened to the real wage for each of these groups since 2007?



In the last 15 years wage inflation and price inflation have indeed been following and correlating with each other, furthermore we can see that in the 2008 recession the wages in fact did surpass price inflation (low educated had even more intense crash to be specific) but they did go back to normal shortly after. moreover its safe to say that the wages of less-educated workers do go throught more extreme changes.

2. [8 Points] Use the WS and PS relations shown above, together with the specification of the function F, to derive a formula for P_t in terms of P_t^e, u_t and parameters α, m, z.

$P_t = (1+m)P_t^e \left(1 - \alpha u_t + z\right)$

3. [8 Points] Consider the equation you have derived in the previous part, and use it to derive the Phillips Curve

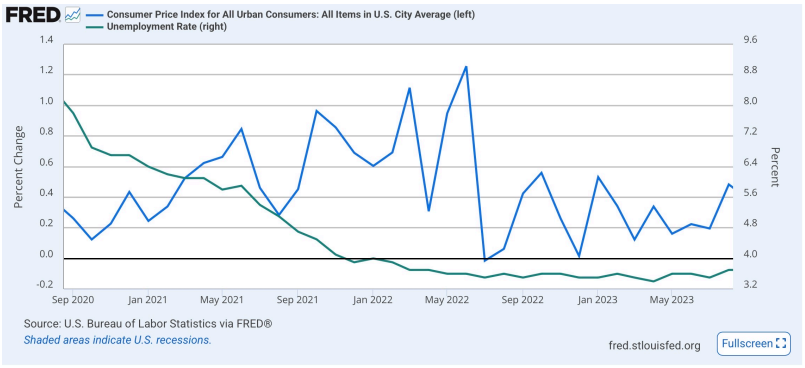
$$\pi_t = \pi_t^e - \alpha\left(u_t - u^n\right),$$

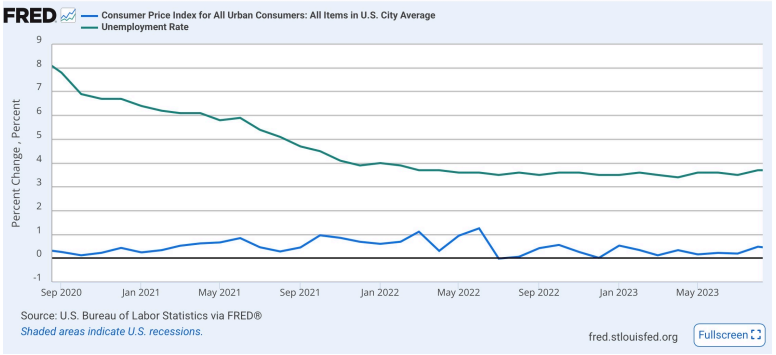
where we have defined the *natural unemployment rate* as $u^n = (m+z)/\alpha$.
Hint: to get started, divide both sides of your equation by P_{t-1}. Then, use the following facts:

$$\frac{P_t}{P_{t-1}} = 1 + \pi_t \qquad \frac{P_t^e}{P_{t-1}} = 1 + \pi_t^e \qquad \frac{1+\pi_t}{\left(1+\pi_t^e\right)(1+m)} \approx 1 + \pi_t - \pi_t^e - m.$$

4. [6 points] Go to the FRED website (link here). Download the data series for the Consumer Price Index for All Urban Consumers: All Items in US City Average (series CPIAUCSL) and the rate of unemployment (UNRATE) of the years 2021 and 2022. Compute monthly inflation in both years and plot the evolution of monthly inflation and the rate of unemployment. What do you observe? Assuming inflation expectations were constant, was the rate of unemployment below or above its natural rate?

I see that CPIAUCSL kept rising while unemployment was falling. Thus with the assumption of $\bar{\pi}$ we can reach the conclusion that $u < u^*$





Question 3 - The IS-LM-PC Model [50 Points]

Consider an economy with aggregate demand,

$$Z_t = c_0 + c_1(Y_t - T) + b_0 + b_1 Y_t - b_2 r_t + G,$$

where t is a time index, Y_t is output, T taxes, r_t the real interest rate, G government spending and $c_0, c_1 b_0, b_1, b_2$ are positive constants so that $c_1 + b_1 < 1$. Next, consider the Phillps curve,

$$\pi_t - \pi_t^e = -\alpha\left(u_t - u_n\right),$$

where $\alpha > 0$ is a positive parameter, u_t is the unemployment rate. The natural rate of unemployment is given by $u_n = \frac{m+z}{\alpha}$ where m is a markup and z is a catch-all variable that stands for all other variables that may affect the natural rate of unemployment.

The labor force $L_t = L$ is constant and the production function is given by

$$Y_t = N_t$$

where the total employment N_t is given by $N_t = (1 - u_t) \cdot L_t$. The variables T, G, m, z, L are all exogenous.

Part (a) Equilibrium of the IS-LM-PC Model [30 Points]

1. [10 Points] Rewrite the Phillips curve relationship in terms of $(Y_t - Y_n)$ instead of $(u_t - u_n)$. Calculate the potential output Y_n in terms of exogenous variables/parameters. (Note: $u_n$is NOT an exogenous variable) $Y_t - Y_n = L - L u_t - L + L u_n = -L(u_t - u_n)$ $u_n = \frac{m+z}{\alpha}$

$u_t - u_n = -\frac{Y_t - Y_n}{L}$ $\pi_t - \pi_t^e = -\alpha\left(-\frac{Y_t - Y_n}{L}\right)$ $Y_n = L(1 - u_n)$

2. [10 Points] Solve for the equilibrium output Y_t as a function of r_t and other exogenous variables/parameters.

$Y_t = c_0 + c_1(Y_t - T) + b_0 + b_1 Y_t - b_2 r_t + G$ $Y_t - c_1 Y_t - b_1 Y_t = c_0 - c_1 T + b_0 - b_2 r_t + G$

$Y_t = \frac{c_0 + c_1 T + b_0 - b_2 r_t + G}{1 - c_1 - b_1}$

3. [10 Points] Using the Phillips curve, solve for the natural rate of interest r_n associated with the medium-run equilibrium, i.e., the interest rate that achieves $Y_t - Y_n = \pi_t - \pi_t^e = 0$. The answer should be in terms of exogenous variables/parameters.

$Y_t = \frac{c_0 + c_1 T + b_0 - b_2 r_t + G}{1 - c_1 - b_1}$ $r_t = \frac{c_0 + c_1 T + b_0 + G - Y_t}{1 - c_1 - b_1} \cdot \frac{1}{b_2}$

Part (b) The Effects of an Increase in the Price of Oil [20 Points]

Remember that the medium run equilibrium is characterized by four conditions:

- Output Y equals its potential level Y_n
- Unemployment rate u equals the natural rate of unemployment u_n
- The real interest rate is equal to the natural rate of interest r_n
- The expected rate of inflation π^e is equal to the actual rate of inflation π.

Using the results in Part (a), answer the following two questions. (provide either a qualitative answer based on economic intuition OR an algebraic derivation using the results in Part (a))

1. [10 Points] Assume $\pi_t^e = \overline{\pi}$, i.e., that inflation expectations are *anchored.* Suppose the economy is initially at the medium-run equilibrium and, suddenly, OPEC succesfully increases oil prices, which can be equivalently seen as an increase in the markup: m increases to m' ($m' > m$). What happens to output and inflation in the new short-run equilibrium? What will the central bank do with the real interest rate to reach the new medium-run equilibrium? What will happen with inflation and output along the transition?

$m \uparrow$ then the gap between inflation and π_t rises thus inflation goes up and also the potencial output decreases.

they will have to increase it.

π will go back to π^e and Y_t will also go to Y_n

2. [10 Points] Suppose we had $\pi_t^e = \pi_{t-1}$ instead. How does this affect your answers to the previous subpoint? What happens with the level of inflation in the new medium-run equilibrium? Explain the intuition.

So in this one even if we get to r^* what happens is that inflation keeps rising thus deflation is needed to go back to normal (2%. for example)

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